

Product-formula defect

$$H = A + B$$

step $S_q(\delta t) \neq U(\delta t)$



Exact residual

$$R_q = U S_q^\dagger = e^{-iK_q}$$

unique, norm-optimal



Local generator

$$K_2 = \delta t^3 K_2^{(3)} + \mathcal{O}(\delta t^5)$$

weight ≤ 3 , local



Learned, transferable

per-site network
train $n = 4, 5 \rightarrow n \leq 10$

stability guarantee: global error $\leq r \eta$ (per-step residual error η , r steps)